INTERCHANGEABILITY

By interchangeable assembly, we mean that identical components, manufactured by different operators, using different machine tools and under different environmental conditions, can be assembled and replaced without any further modification during the assembly stage and without affecting the functioning of the component when assembled.

Production on an interchangeable basis results in an increased productivity with a corresponding reduction in manufacturing cost. When components are produced in bulk, unless they are interchangeable, the purpose of mass production is not fulfilled.

SELECTIVE ASSEMBLY

A product's performance is often influenced by the clearance or, in some cases, by the preload of its mating parts. Achieving consistent and correct clearances and preloads can be a challenge for assemblers.

Tight tolerances often increase assembly costs because labour expenses and the scrap rate go up. The tighter the tolerances, the more difficult and costly the component parts are to assemble. Keeping costs down while maintaining tight assembly tolerances can be made easier by a process called selective assembly, or match gauging.

The term selective assembly describes any technique used when components are assembled from subcomponents such that the final assembly satisfies higher tolerance specifications than those used to make its subcomponents. The use of selective assembly is inconsistent with the notion of interchangeable parts, and the technique is rarely used at this time.

To match gauge for selective assembly, one group of components is measured and sorted into groups by dimension, prior to the assembly process. This is done for both mating parts. One or more components are then measured and matched with a presorted part to obtain an optimal fit to complete the assembly. It results in complete protection against defective assemblies and reduces the matching cost.

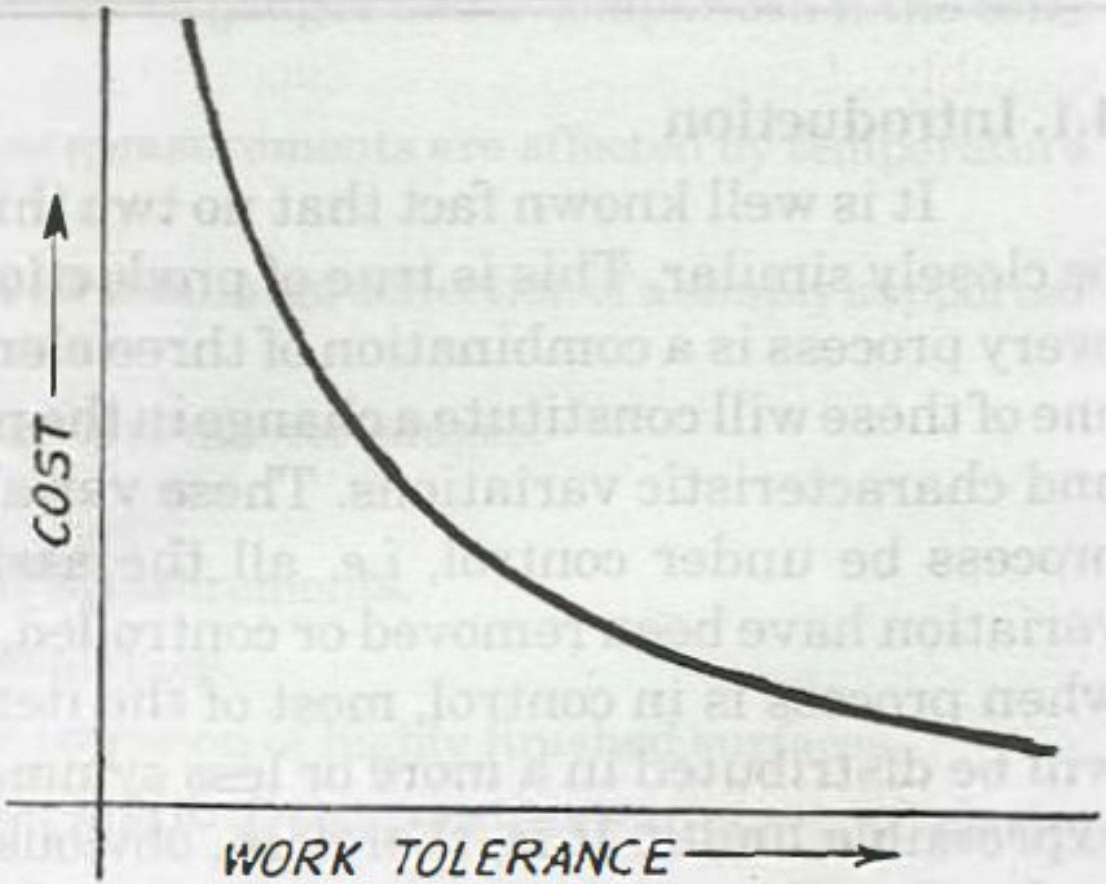
Limits , Fits , Tolerances

In nature two extremely similar (identical) things are difficult to obtain. If at all we come across exactly similar things, it must be only by chance. This fact holds good for production of component parts in engineering also. No production process is good enough to produce all items of products exactly alike.

1. It is not possible to make any part precisely to a given dimension, due to variability of elements of production processes.

2. Secondly, even if by chance the part is made exactly to a given dimension, it is impossible to measure it accurately enough to prove it.

3. Thirdly, if attempts are made to achieve perfect size the cost of production will increase tremendously.


COST
WORK TOLERANCE

LIMITS

In mass production, where large number of parts are to be made by different operators on different machines, it is impossible to make all parts exactly alike and to exact dimensions. The difference in dimension does exist because of these variables. Also, perfect size is not only difficult but a costly affair.

It is, therefore, obvious that some permissible variation jn dimension has to be allowed to account for variability. The dimension of the manufactured part can thus only be made to lie between two limits, maximum and minimum.

Therefore, the ranges of permissible difference in dimension have been standardized under the name limits. The limits of size of a dimension of a part are two extreme permissible sizes, between which the actual size of the dimension may lie. They are fixed with reference to the basic size of that dimension. The high limit (upper limit) for that dimension is the largest size permitted and the low limit is the smallest size permitted for that dimension.

TOLERANCE

The permissible variation in size or dimension is called tolerance.

Thus, the word tolerance indicates that a worker is not expected to produce the part to the exact size, but a definite small size error is permitted.

The difference between the upper limit (high limit) and the lower limit of a dimension represents the margin for variation in workmanship, and is called a '**Tolerance Zone**'.

A shaft of 25 mm basic size may be written as 25 ± 0.02.

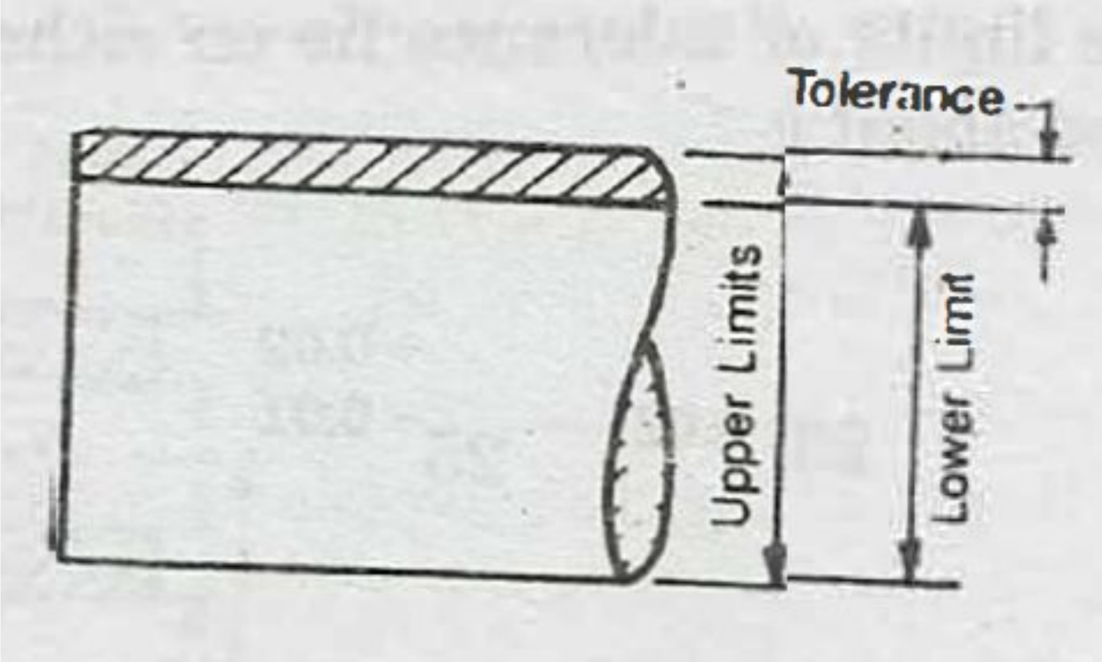
The mnmum permissible size (upper limit)= 25.02 mm

and the minimum permissible size (lower limit)= 24.98 mm

Then,

Tolerance = Upper limit- Lower limit

= 25.02 - 24.98 = 0.04 mm.



Systems of Writing Tolerances

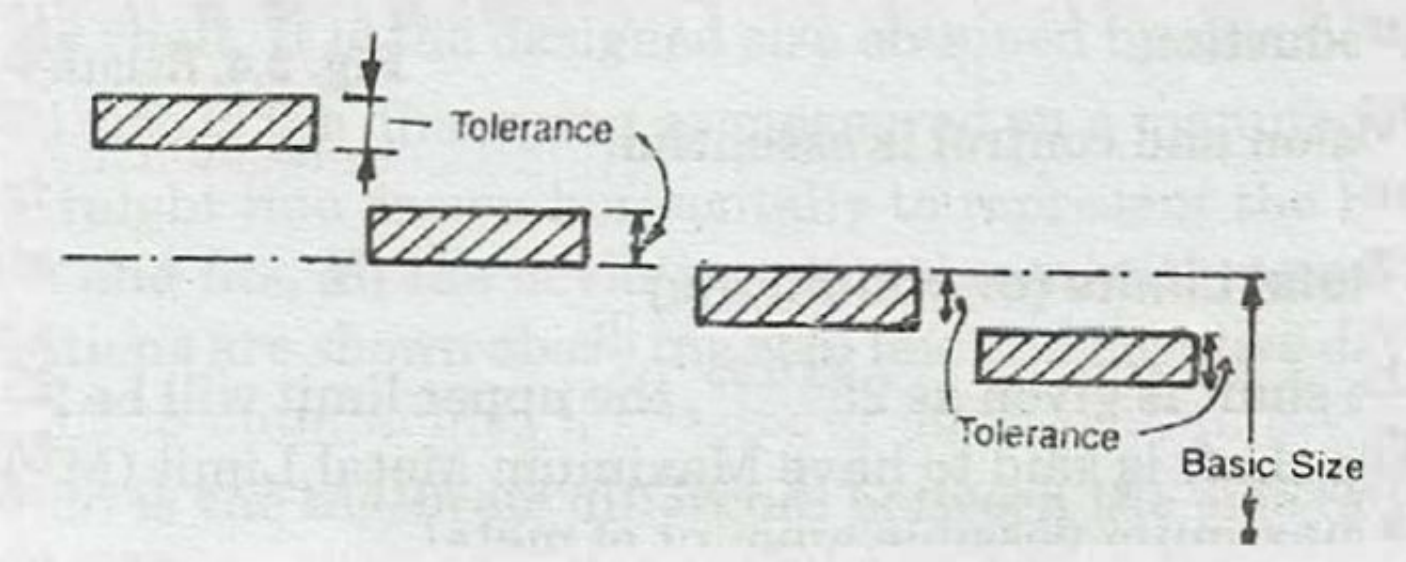
There are two systems of writing tolerances:

(i) Unilateral system

(ii) Bilateral system

Unilateral System

In this system, the dimension of a part is allowed to vary only on one side of the basic size i.e.,tolerance lies wholly on one side of the basic size either above or below it.



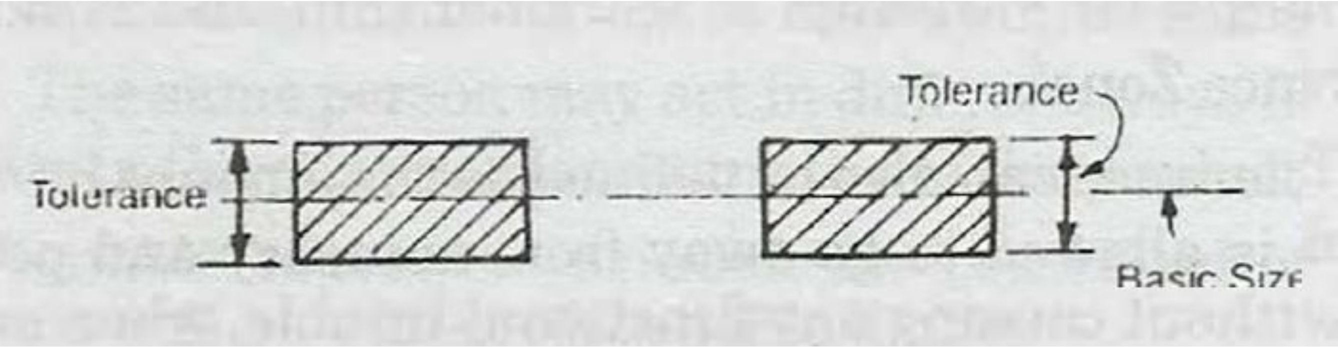
Examples of unilateral tolerance are

$$25^{+0.02}_{+0.01},\quad 25^{+0.02}_{-0.00},\quad 25^{-0.01}_{-0.02},\quad 25^{+0.00}_{-0.02}\ \text{etc.}$$

Bilateral System

In this system, the dimension of the part is allowed to vary on both the sides of the basic size i.e., the limits of tolerance lie on either side of the basic size; but may not be necessarily equally disposed about it.

$25^{\pm 0.02}$, $25^{+0.02}_{-0.01}$



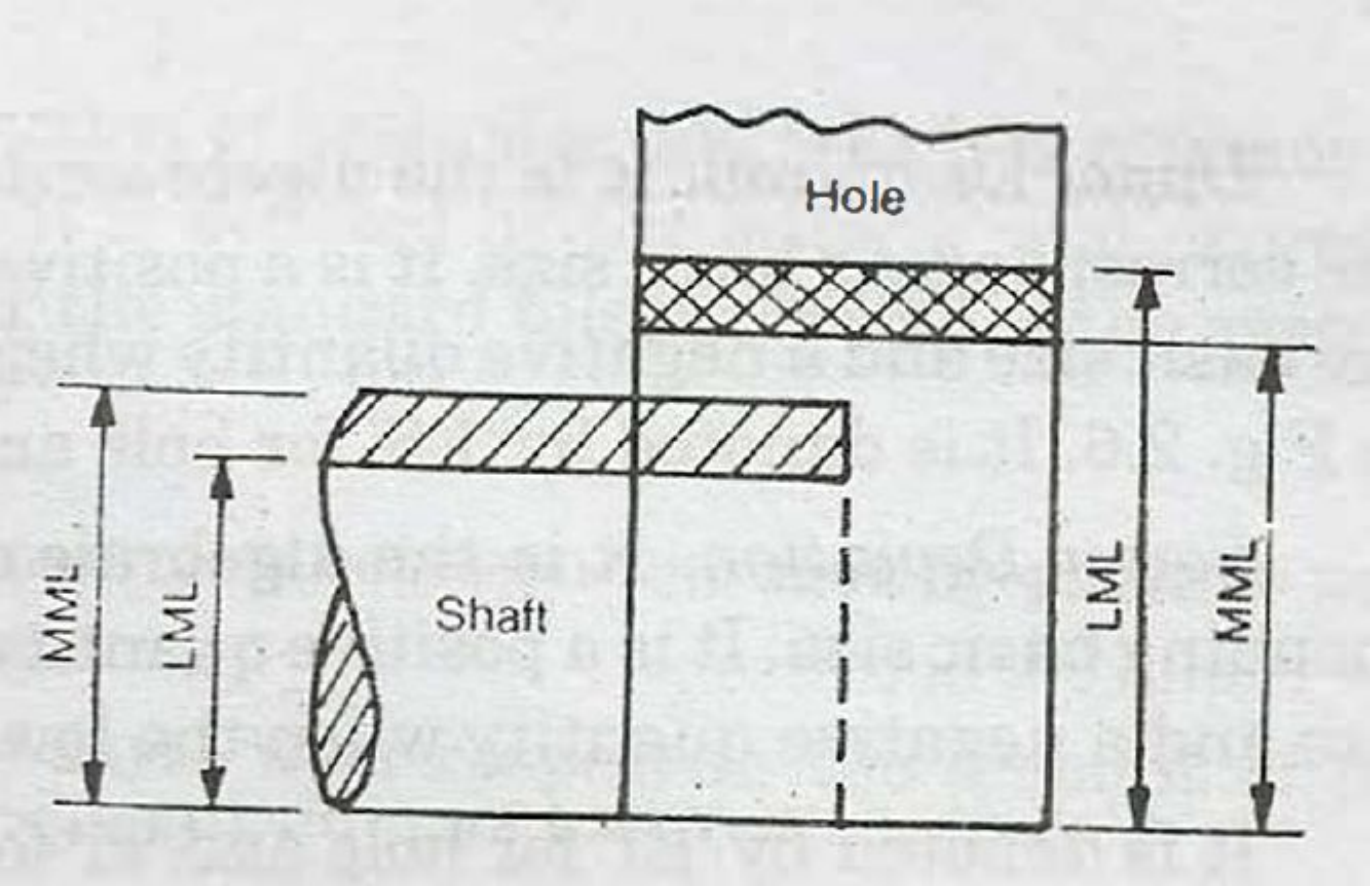
Maximum and Minimum Metal Limits (or conditions)

If the tolerance for the shaft is given as 25± 0·05, the upper limit will be 25. 05 mm and the lower limit will be 24.94 mm.

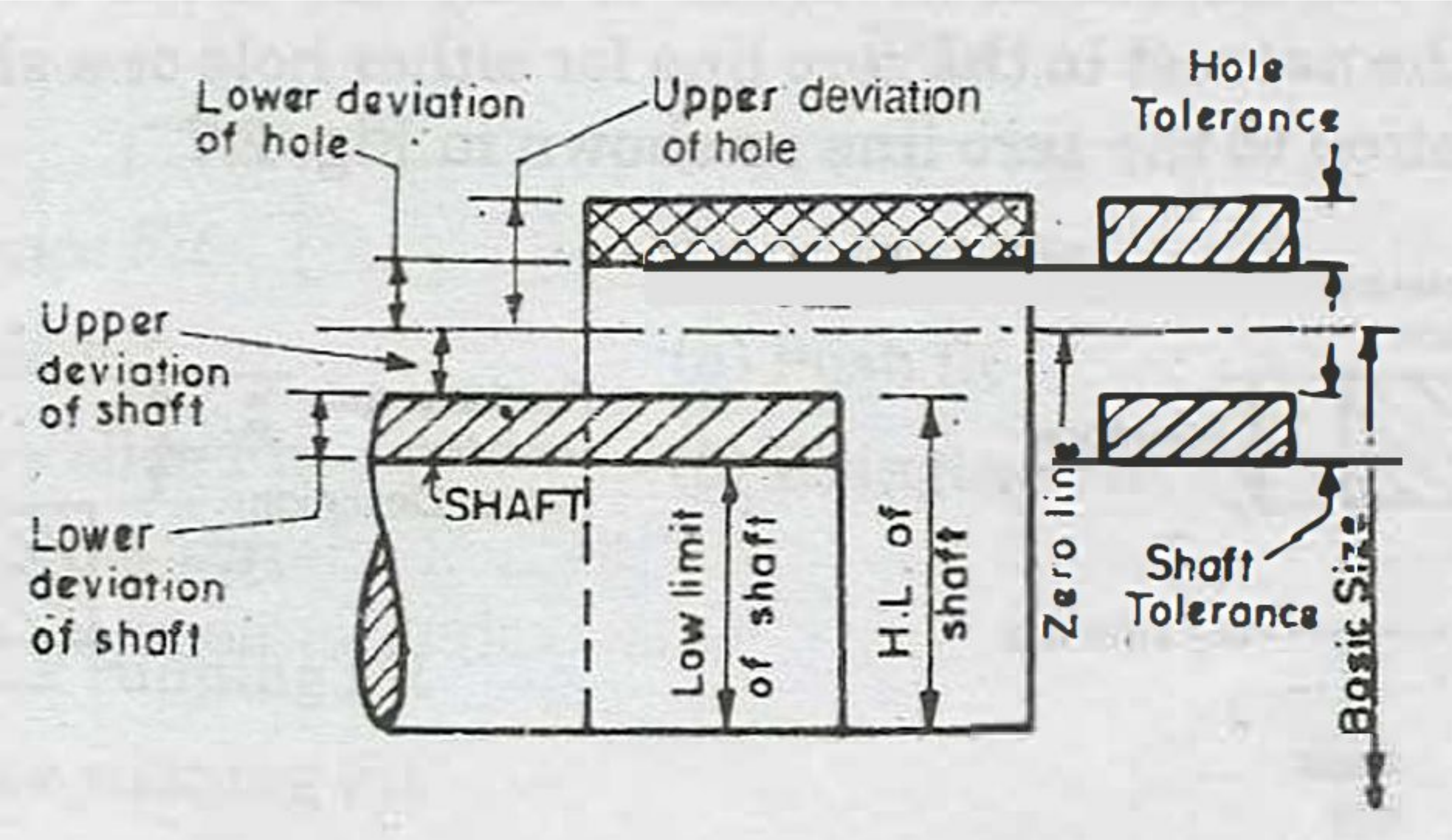
The shaft is said to have Maximum Metal Limit (MML) of 25.05 mm, since at this limit the shaft has maximum possible amount of metal.

The limit of 24.95 will then be the minimum or "Least Metal Limit" (LML) because at this limit the shaft will have the least possible amount of metal.

Similarly, if the hole is designated as 3o+-o.o5 mm, the upper limit will be 30.05 mm and the lower limit will be 29.95 mm. Then, the Maximum Metal Limit (MML) of hole will be equal to 29.95, since at this lower limit the hole has the maximum possible amount of metal; while the upper limit of 30.05 mm will be the minimum of 'Least Metal Limit' (LML) of hole as, at this limit the hole will have the least possible amount of metal.


Hole
Shaft
MML
LML
LML
MML

Conventional Diagram of Limits and Fits



Terminology for Limits and Fits

Shaft : The term shaft refers not only to the diameter of a circular shaft but also to any external dimension of a component.

Hole. The term hole refers not only to the. diameter of circular hole but also to any internal dimension of a component.

When an assembly is made of two parts, one is known as male-surface and the other mating part as a female (enveloping) surface. The male surface is called as 'Shaft' and the female surface as 'Hole'.

Basic or Nominal Size. It is the standard size of a part with reference to which the limits of variation of a size are determined. The basic size is the same for the hole and its shaft. It is the designed size obtained by calculations for strength.

Actual Size. Actual size is the dimension as measured on a manufactured part.

Zero line. It is a straight line drawn horizontally to represent the basic size. In the graphical representation of limits and fits, all the deviations are shown with respect to the zero line (datum line). The positive deviations are shown above the zero line and negative deviations below.

Deviation. Deviation is the algebraic difference between the size (actual, maximum etc.) and the corresponding basic size.

Upper Deviation. It is the algebraic difference between the upper (maximum) limit of size and the corresponding basic size. It is a positive quantity when the maximum limit of size is greater than the basic size and a negative quantity when the upper limit of size is less than the basic size as shown in Fig.

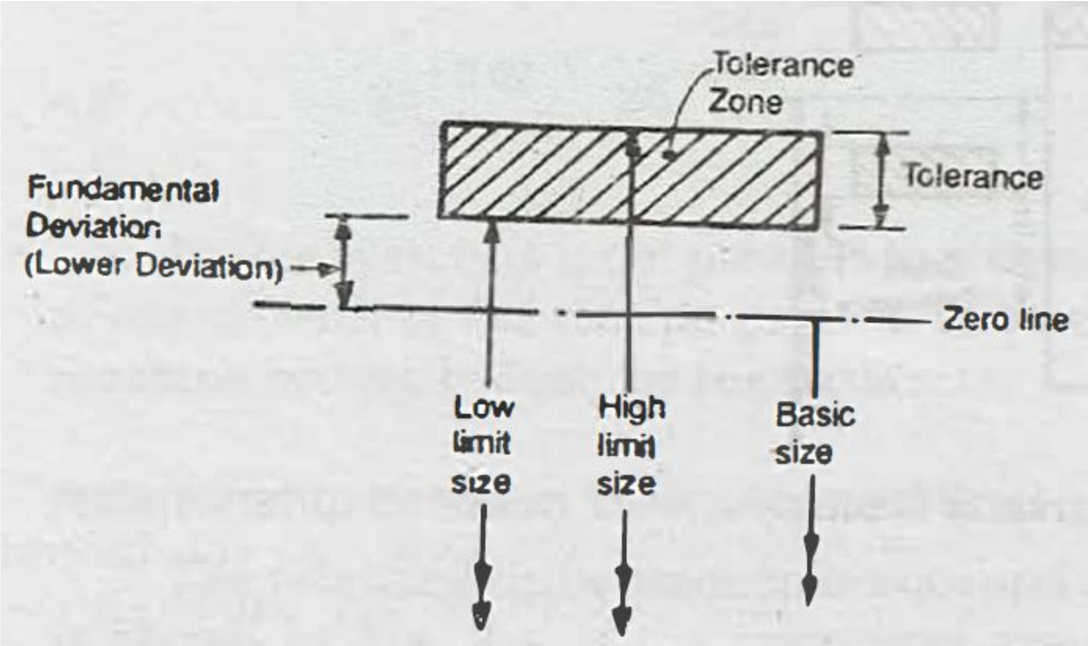
It is denoted by 'ES' for hole and 'es' for a shaft.

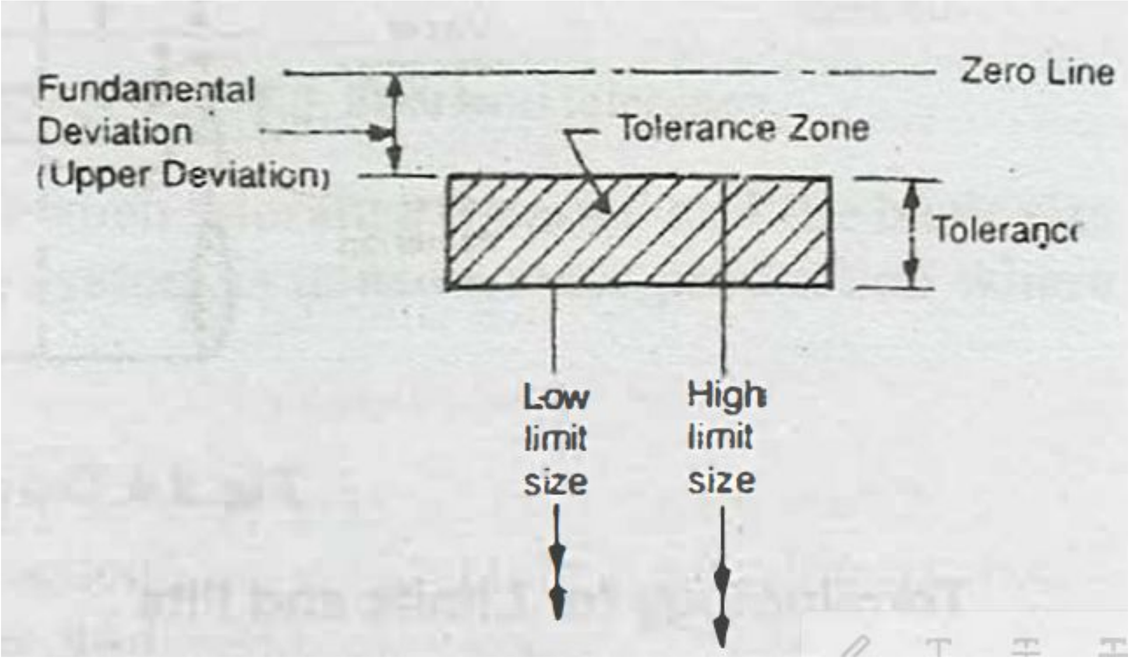
Lower Deviation. It is the algebraic difference between the lower limit of size and the corresponding basic size. It is a positive quantity when the minimum limit of size is greater than the basic size and a negative quantity when the lower limit of size is less than the basic size .

It is denoted by 'EI' for hole and 'ei' for shaft

Fundamental Deviation

Fundamental deviation is that one of the two deviations (either the upper or the lower) which is the nearest to the zero line for either hole or a shaft. It fixes the position of the 'Tolerance Zone' in relation to the zero line as shown in Fig.





Lower deviation as fundamental deviation

Upper deviation as fundamental deviation

The fundamental deviation (25 nos) for the hole is denoted by capital letters A, B, C, ... ZC and the same for shaft is denoted by small letters a, b, c ..... zc etc.

excluding I,L,O,Q & W and four sets of letters JS, ZA, ZB & ZC are included.

From Fig. it is clear that when the tolerance zone is above the zero line, lower deviation is the fundamental deviation. While, when the tolerance zone is below the zero line, upper deviation is the fundamental deviation.

Basic shaft. Basic shaft is the shaft whose upper deviation is zero. Thus the upper limit of the basic shaft is the same as the basic size. It is denoted by letter .'h'.

Basic Hole. Basic hole is that hole whose lower deviation is zero i.e., its low limit is the same as the basic size. It is denoted by a letter 'H'.

Tolerance Zone. It is the zone bounded by two limits of size of a part in the graphical representation of tolerance. It is defined by its magnitude and by its position in relation to the zero line as shown in Fig.

Tolerance grade. The tolerance grade is an indication of the degree of accuracy of manufacture and it is designated by the letters 'IT' followed by a number, where 'IT ‘ stands for "International Tolerance grade". Tolerance grades are IT01, IT0, ITl, up to IT16, the larger the number larger will be the tolerance.(18 grades)

IS 919 (Part 2) : 1993
ISO 286-2 : 1988

Indian Standard

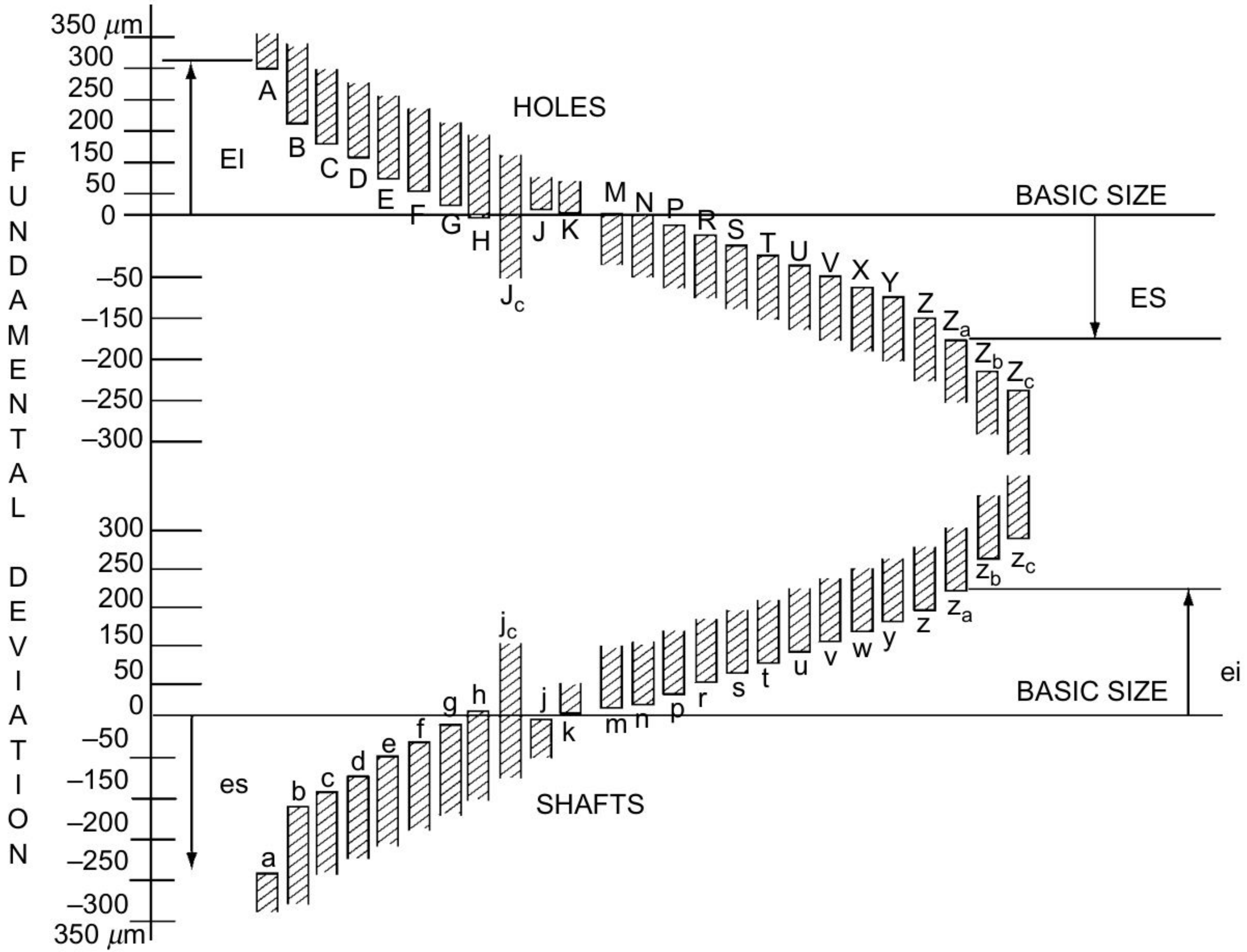
ISO SYSTEM OF LIMITS AND FITS

PART 2 TABLES OF STANDARD TOLERANCE GRADES AND LIMIT DEVIATIONS FOR HOLES AND SHAFTS

BIS IS 696 : 1972
CODE OF PRACTICE FOR GENERAL ENGINEERING DRAWINGS

Standard Tolerance Unit.

A unit which is a function of basic size and which is common to the formula defining the different grades of tolerances. It is denoted by the letter 'i' and expressed in terms of microns. It serves as a basis for determining the standard tolerance (IT) of the system.



FUNDAMENTAL DEVIATION
350 μm
300
250
200
150
50
0
–50
–150
–200
–250
–300
EI
A
B
C
D
E
F
G
H
J_c
J
K
M
N
P
R
S
T
U
V
X
Y
Z
Z_a
Z_b
Z_c
HOLES
BASIC SIZE
ES
300
250
200
150
50
0
–50
–150
–200
–250
–300
350 μm
es
a
b
c
d
e
f
g
h
j_c
j
k
m
n
p
r
s
t
u
v
w
y
z
z_a
z_b
z_c
SHAFTS
BASIC SIZE
ei

FIT

Fit may be defined as a degree of tightness or looseness between two mating parts to perform a definite function when they are assembled together.

It is the relationship between the two mating parts with respect to the amount of play or tightness which is present when they are assembled together.

Accordingly, a fit may result either in a movable joint or a fixed joint. For example, a shaft running in a bearing can move in relation to it and thus forms a movable joint, whereas, a pulley mounted on the shaft forms a fixed joint.

Types of Fits (Classification of Fits)

On the basis of positive, zero and negative values of Clearance, there are three basic types of fits:

1) Clearance Fit (2) Transition Fit and, (3) Interference Fit.

These are further classified in the following manner :

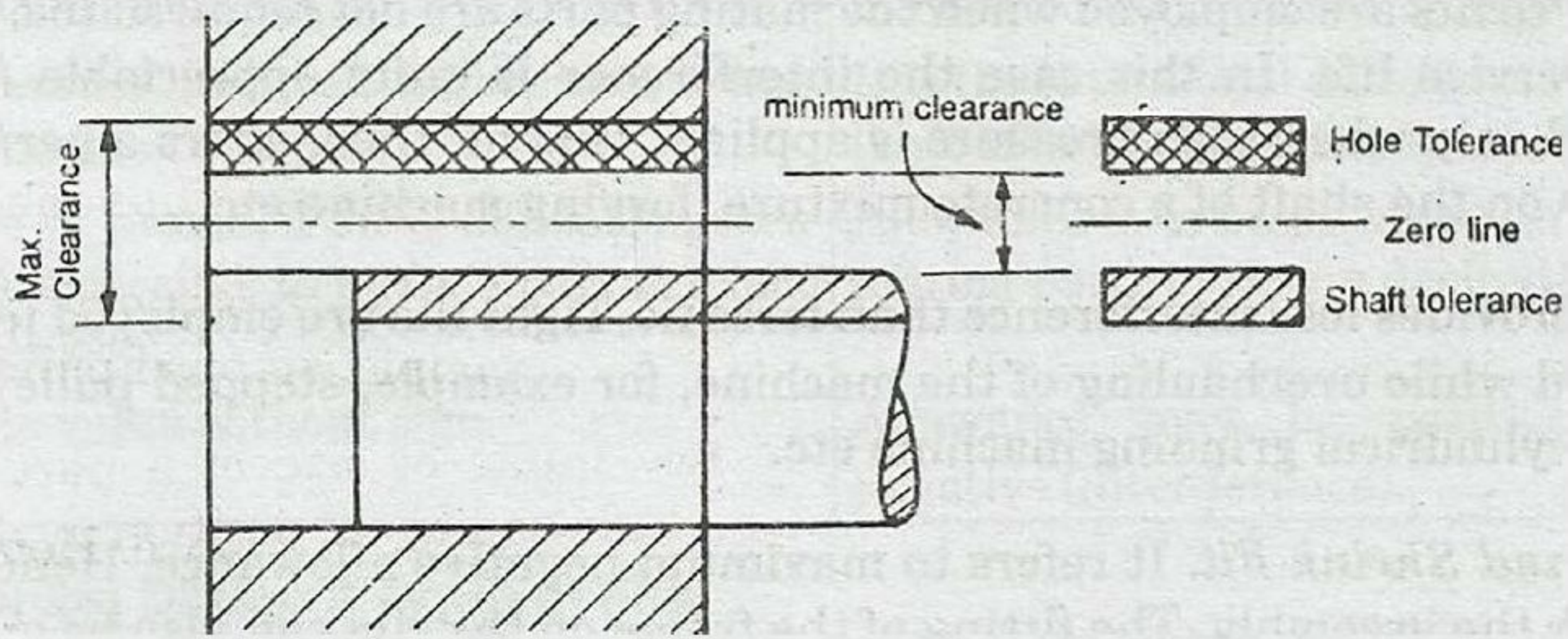
Fits

- *Clearance Fit*
  - (*a*) Slide Fit
  - (*b*) Easy slide Fit
  - (*c*) Running Fit
  - (*d*) Slack running Fit
  - (*e*) Loose running Fit
- *Transition Fit*
  - (*a*) Push fit
  - (*b*) Wringing Fit
- *Interference Fit*
  - (*a*) Force Fit
  - (*b*) Tight Fit
  - (*c*) Shrink fit

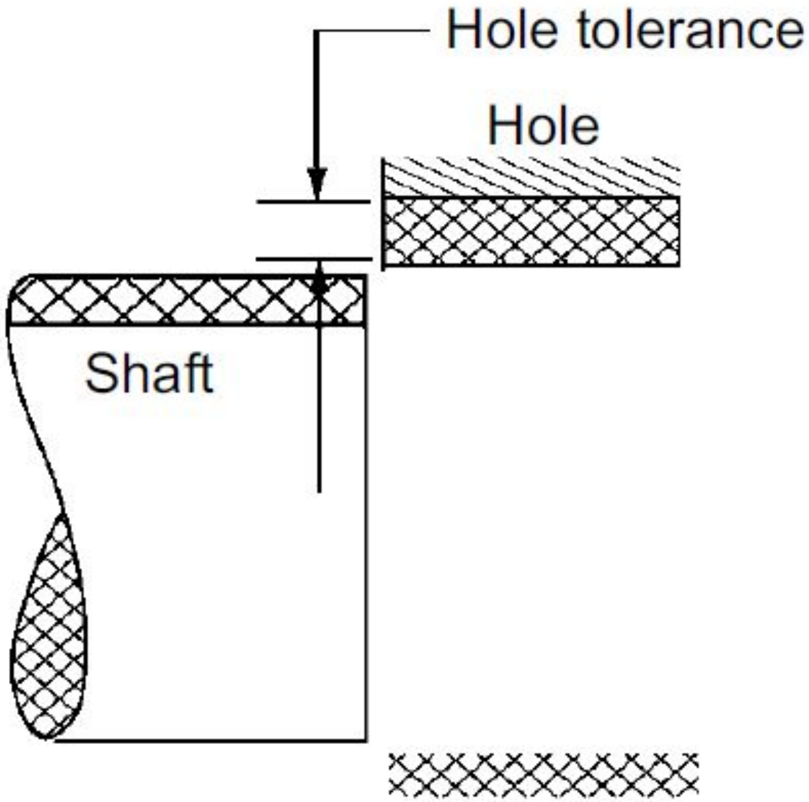
Clearance Fit

In this type of fit shaft is always smaller than the hole i.e., the largest permissible shaft diameter is smaller than the diameter of smallest hole. So that the shaft can rotate or slide through with different degrees of freedom according to the purpose of mating part.

Clearance fit exists when the shaft and the hole are at their maximum metal conditions. The tolerance zone of the hole is above that of the shaft as shown in Fig.



Max.
Clearance
minimum clearance
Hole Tolerance
Zero line
Shaft tolerance


Hole tolerance
Hole
Shaft

Maximum Clearance.

It is the difference between the minimum size of shaft and maximum size of hole.

Minimum Clearance.

It is the difference between the maximum size of shaft and minimum size of hole.

1. Slide Fit. This type of fit has a very small clearance, the minimum clearance being zero. Sliding fits are employed when the mating parts are required to move slowly in relation to each other e.g., tailstock spindle of lathe, feed movement of the spindle quill in a drilling machine, sliding change gears in quick change gear box of a centre lathe etc.

2. Easy Slide Fit. This type of fit provides for a small guaranteed clearance. It serves to ensure alignment between the shaft and hole. It is applicable for slow and non-regular motion, for example, spindle of lathe and dividing heads, piston and slide valves, spigots etc.

3. Running Fit. Running fit is obtained when there is an appreciable clearance between the mating parts. The clearance provides a sufficient space for a lubrication film between mating friction surfaces. It is employed for rotation at moderate speed, e.g., grear box bearings, shaft pulleys, crank shafts in their main bearings etc.

4. Slack running Fit. It is obtained when there is a considerable clearance between the mating parts. This type of fit may be required as compensation for mounting errors e.g., cam shaft of IC. engine, shaft of certifigual pump etc.

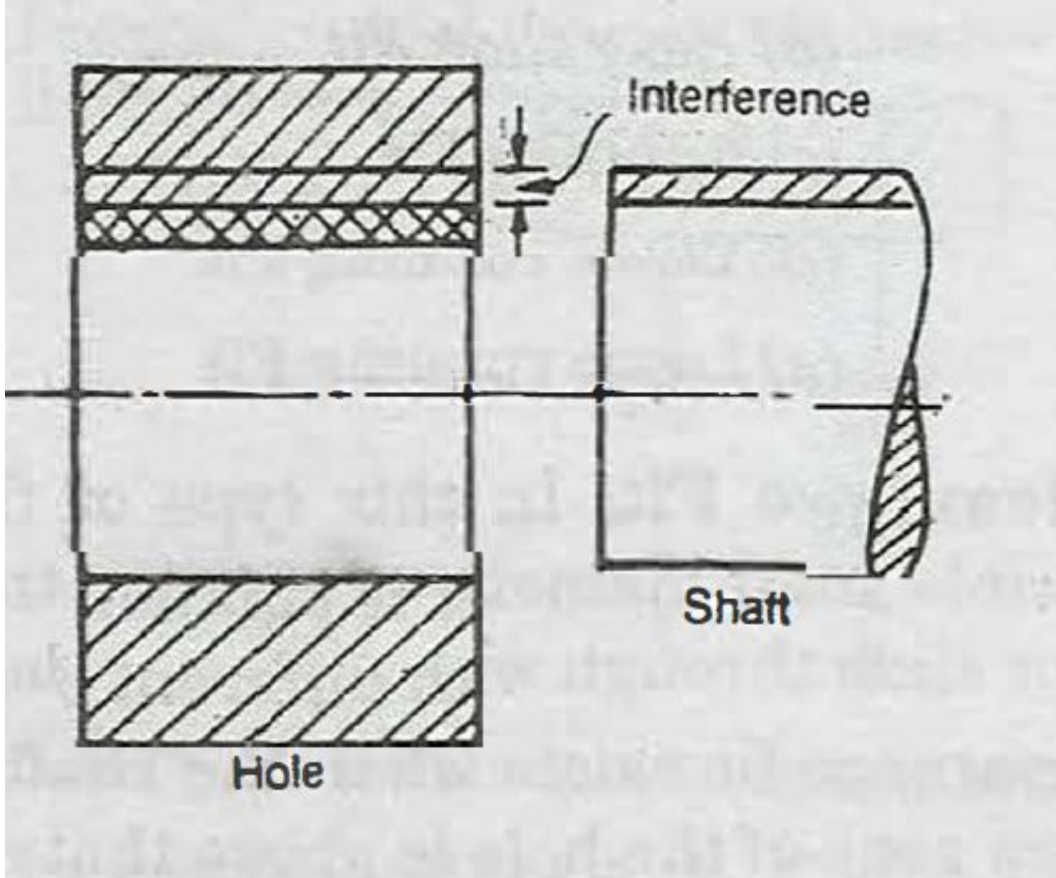
5. Loose running Fit. Loose running fit is employed for rotation at very high speed, e.g., idle pulley on their shaft such as that used in quick return mechanism of a planer.

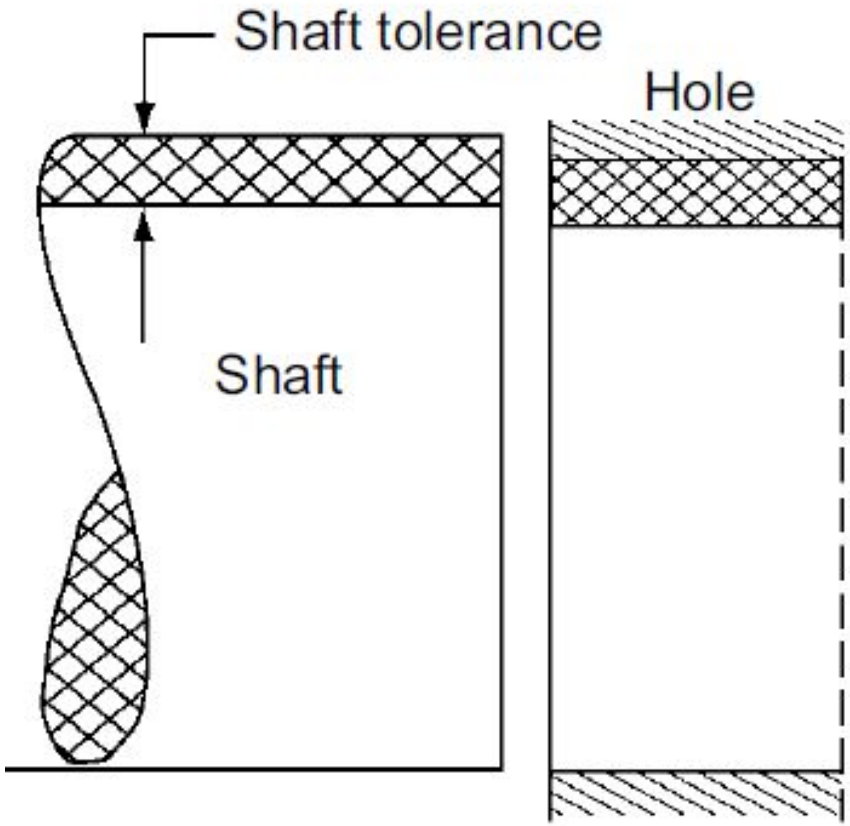
Interference Fit

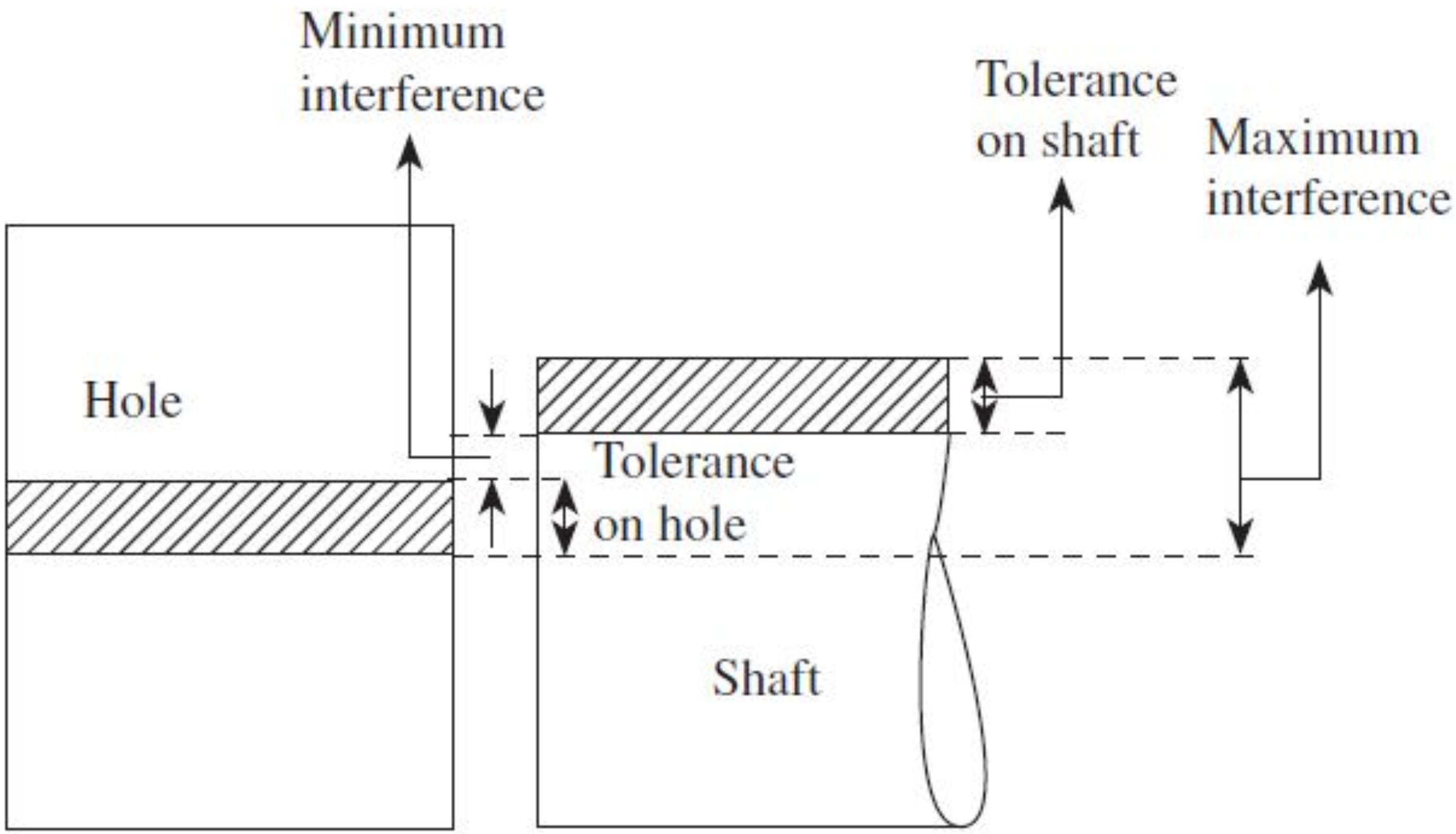
In this type of fit the minimum permissible diameter of the shaft is larger than the maximum allowable diameter of the hole.

Thus the shaft and the hole members are intended to be attached permanently and used as a solid component.

Interference fit




Shaft tolerance
Hole
Shaft


Minimum
interference
Tolerance
on shaft
Maximum
interference
Hole
Tolerance
on hole
Shaft

Minimum Interference

In case of interference fit, it is the arithmetical difference between maximum size of the hole and the minimum size of shaft before assembly.

Maximum Interference

In case of interference fit, it is the arithmetical difference between the minimum size of the hole and the maximum size of the shaft before assembly.

Elastic strains developed on the mating surfaces during the process of assembly prevent relative movement ofthe mating parts.

For example, steel tyres on railway car wheels, gears on intermediate shafts of trucks, bearing in the gear of a lathe head stock, drill bush in jig plate,cylinder linear in block, steel rings on a wooden bullock cart wheels etc.

1. Force Fit. Force fits are employed when the mating parts are not required to be disassembled during their total service life. In this case the interference is quite appreciable and, therefore, assembly is obtained only when high pressure is applied. This fit, thus, offers a permanent type of assembly, e.g., gears on the shaft of a concrete mixture, forging machine etc.

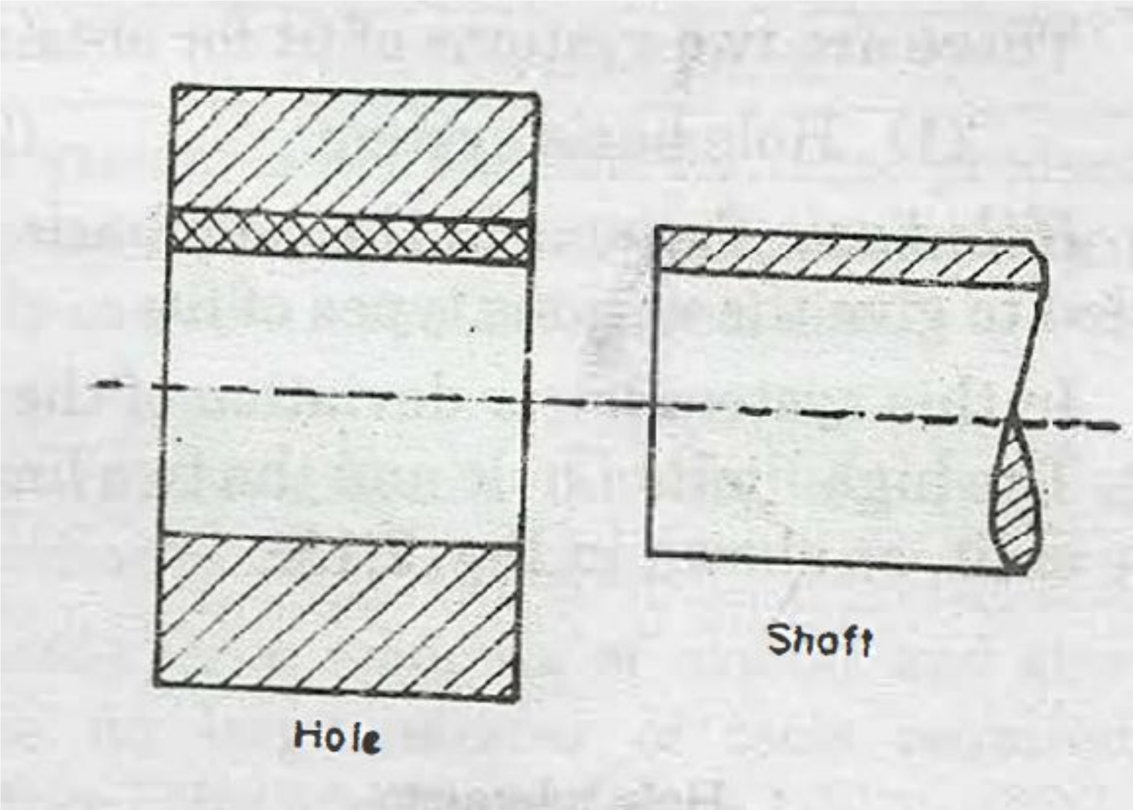
2. Tight Fit. It provides less interference than force fit. Tight fits are employed for mating parts that may be replaced while overhauling of the machine, for example, stepped pulleys on the drive shaft of a conveyor, cylindrical grinding machine etc.

3. Heavy force and Shrink Fit. It refers to maximum negative allowance. Hence considerable force is necessary for the assembly. The fitting of the frame on the rim can also be obtained first by heating the frame and then rapidly cooling it in its position.

Transition Fit

Transition fit lies mid way between clearance and interference fit. In this type the size limits of mating parts (shaft and hole) are so selected that either clearance- or indifference may occur depending upon the actual sizes of the parts. Push fit and wringing fit are the examples of this type of fit.

Transition fit



In this type of fit the tolerance zones of the bole and shaft overlap completely or in part.

1. Wringing Fit. A wringing fit provides either zero interference or a clearance. These are used where parts can be replaced without difficulty during minor repairs.

2. Push Fit. The fit provides small clearance. It is employed for parts that must be dis-assembled during operation of a machine for example, change gears, slip bushing etc.

Allowance

Allowance is the prescribed difference between the dimensions of two mating parts for any type of fit.

It is defined as the difference between maximum material limit of shaft and hole.

OR

It is the intentional difference between the lower limit of hole and higher limit of the shaft.

Depending upon the fit , it is equal to minimum clearence or maximum interference.

The allowance may be positive or negative.

The positive allowance is called clearance and the negative allowance is called interference.

Difference between Tolerance and Allowance

Tolerance	*Allowance*
1. It is the permissible variation in dimension of a part (either a hole or a shaft).	It is the prescribed difference between the dimensions of two mating parts (hole and shaft).
2. It is the difference between higher and lower limits of a dimension of a part.	It is the intentional difference between the lower limit of hole and higher limit of shaft.
3. The tolerance is provided on a dimension of a part as it is not possible to make a part to exact specified dimension.	Allowance is to be provided on the dimension of mating parts to obtain desired type of fit.
4. It has absolute value without sign.	Allowance may be positive (clearance) or negative (inter-ference).

Systems of Obtaining Different Types of Fits

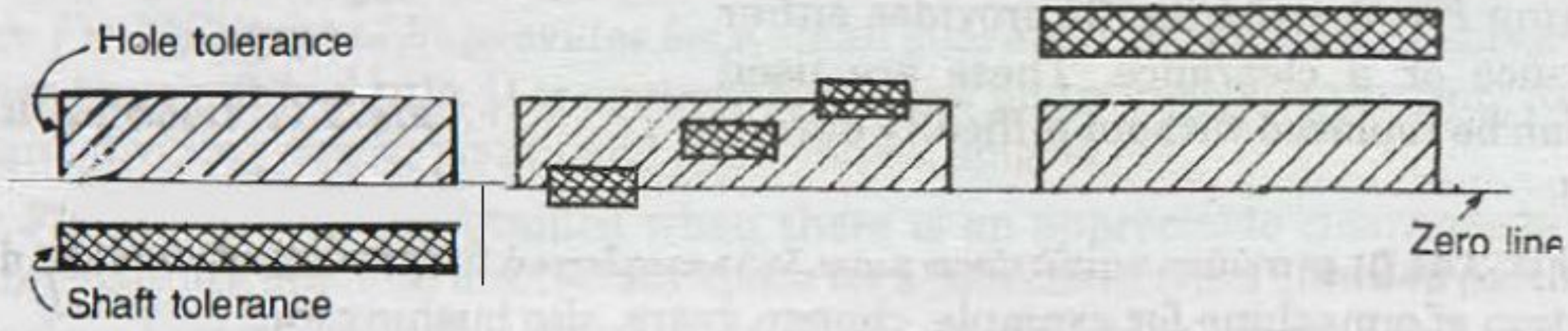
(1) Hole basis system

(2) Shaft basis system

Hole basis System

In the hole basis system the hole is kept constant and the shaft sizes are varied to give the various types of fits.

In this system lower deviation of the hole is zero i.e., the low limit of hole is the same as basic size. The high limit of hole and the two limits of size for the shaft are then varied to give the desired type of fit, as shown in Fig.

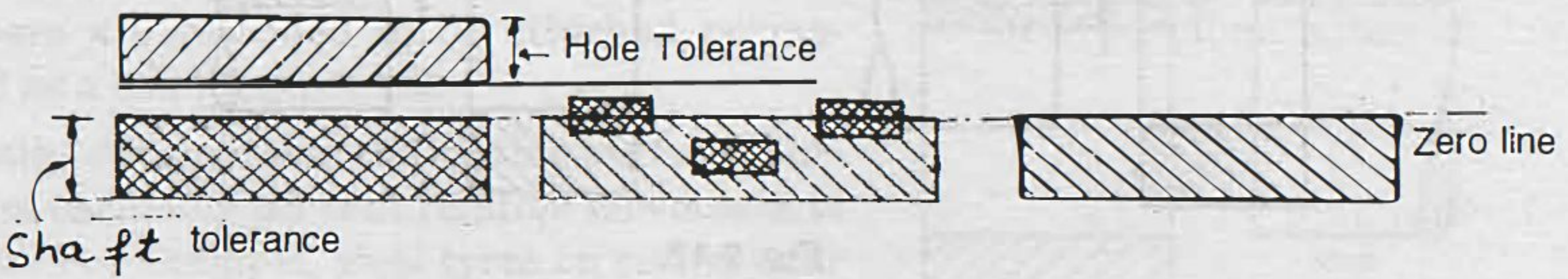


(a) Clearance Fit (b) Transition Fit (c) Interference Fit

Shaft basis System

In the shaft basis system the shaft is kept constant and the sizes of the hole are varied to give various types of fits.

In this system the upper deviation (fundamental deviation) of shaft is zero i.e., the high limit of shaft is the same as basic size and the various fits are obtained by varying the low limit of shaft and both the limits of hole.



(a) Clearance Fit (b) Transition Fit (c) Interference Fit

The hole basis system is most commonly used because it is more convenient to make correct holes of fixed sizes, since the standard drills, taps, reamers and broaches etc. are available for producing holes and their sizes are not adjustable. On the other hand size of shaft produced by turning, grinding etc. can be very easily varied.

Difference between 'Hole Basis' and 'Shaft Basis' Systems

Hole Basis System	*Shaft Basis System*
1. Size of hole whose lower deviation is zero (H-hole) is assumed as the basic size.	Size of shaft whose upper deviation is zero (*h*-shaft) is assumed as basic size.
2. Limits on the hole are kept constant and those of shaft are varied to obtain desired type of fit.	Limits on the shaft are kept constant and those on the hole are varied to have necessary fit.
3. Hole basis system is preferred in mass production, because it is convenient and less costly to make a hole of correct size due to availability of standard drills and reamers.	This system is not suitable for mass production because it is convenient, time consuming and costly to make a shaft of correct size.
4. It is much more easy to vary the shaft sizes according to the fit required.	It is rather difficult to vary the hole sizes according to the fit required.
5. It requires less amount of capital and storage space for tools needed to produce shafts of different sizes.	It needs large amounts of capital and storage space for large number of tools required to produce holes of different sizes.
6. Gauging of shafts can be easily and conveniently done with adjustable gap gauges.	Being internal measurement, gauging of holes cannot be easily and conveniently done.

In a limit system, the following limits are specified for a hole and shaft assembly:

$$\text{Hole} = 30^{+0.02}_{+0.00}\ \text{mm and shaft} = 30^{-0.02}_{-0.05}\ \text{mm}$$

Determine the (a) tolerance and (b) allowance.

Tolerance for a hold and shaft assembly having a nominal size of 40 mm are as follows:

$$\textbf{Hole} = 40^{+0.06}_{+0.02}\,\text{mm}\ \textbf{and shaft} = 40^{-0.06}_{-0.08}\,\text{mm}$$

Determine MML of hole

The following limits are specified in a limit system, to give a clearance fit between a hole and a shaft:

$$\text{Hole} = 25^{+0.02}_{0-.01}\ \text{mm and shaft} = 25^{-0.004}_{-0.02}\ \text{mm}$$

Determine tolerance on shaft.

2) Find out the type of fit for hole $40^{+0.020}_{+0.000}$ and shaft $40^{+0.010}_{-0.010}$

In an interchangeable assembly, shafts of size $25.000^{+0.040}_{-0.010}$ mm mate with holes of size $25.000^{+0.030}_{+0.020}$ mm. The maximum interference (in microns) in the assembly is

INDIAN STANDARDS SPECIFICATIONS AND APPLICATION (**IS-919**)

The Indian standards are in line with the ISO (International Organisations for Standards) recommendations.

It consists of suitable combinations of 18 grades of fundamental tolerances or in other words grades of accuracy for manufacture, and 25 types of fundamental deviations.

The 18 grades of fundamental tolerances are designated as ITO1, ITO, IT1 to IT16. While, the fundamental deviations are indicated by letter symbols for both hole and shaft (capital letters “A to Zc” for holes and small letters “a to Zc “ for shafts. These are : A, B, C, D, E, F, G, H, J_s, J, K, M, N, P, R, S, T, U, V, X, Y, Z, Z_A, Z_B, Z_C).

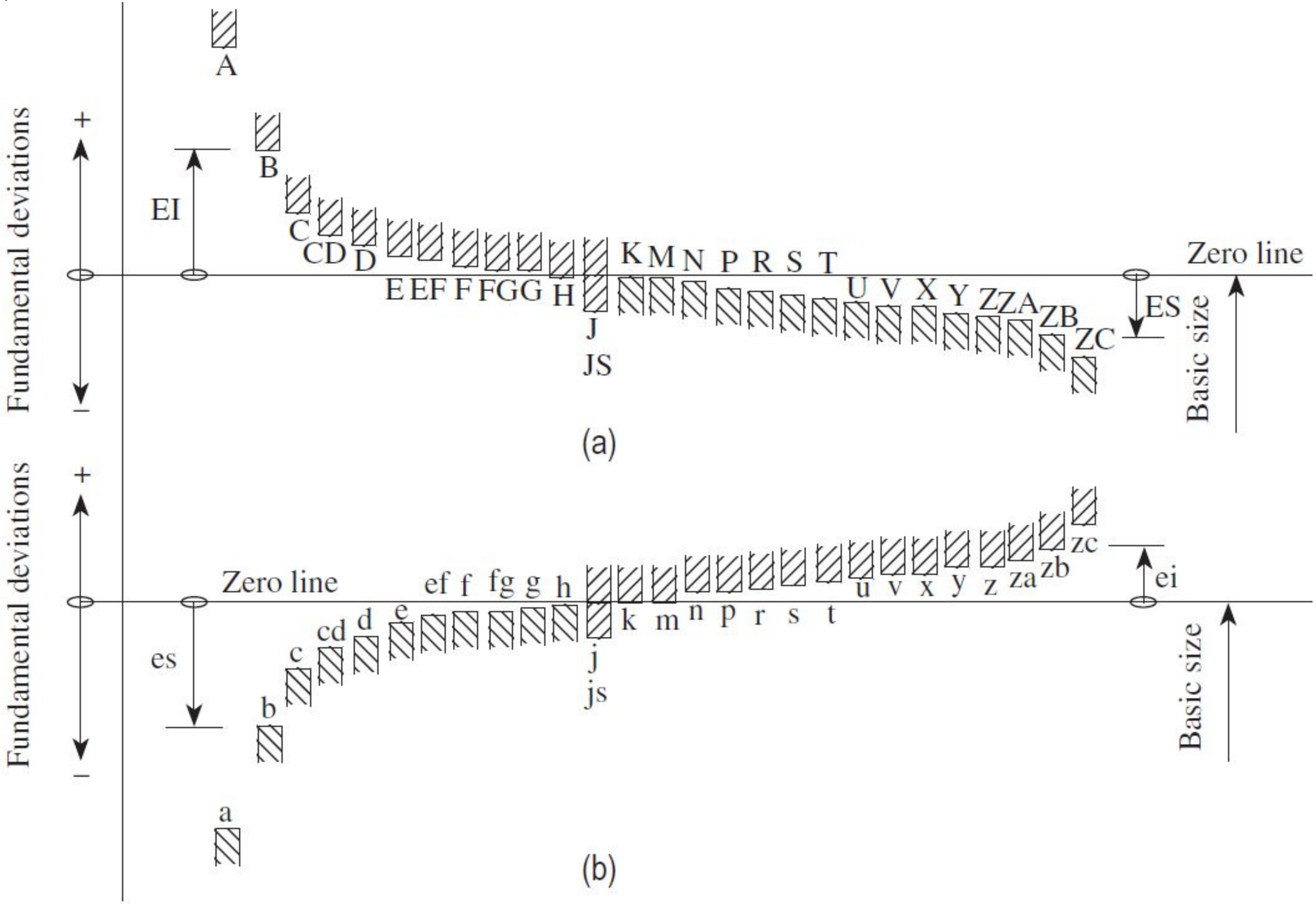
There is no ” I , L , O , Q , W “

Innumerable fits ranging from extreme clearance to those of extreme interference can be obtained by a suitable combination of fundamental tolerances and fundamental deviations. Each of 25 holes has a choice of 18 tolerances.

For shafts 'a' to 'h' the upper deviation is below the zero line and for shafts 'j' to 'Zc' it is above the zero line.

For holes 'A' to 'H' lower deviation is above the zero line and for I to Zc it is below the zero line as shown in Fig.

Typical representation of different types of fundamental deviations (a) Holes (internal features) (b) Shafts (external features)



The algebraic relation is :

$$\left.\begin{aligned} ei &= es - IT \\ es &= ei + IT \end{aligned}\right\} \quad \text{for shaft,}$$

The ISO system provides tolerance grades from IT01, IT0, and IT1 to IT16 to realize the required accuracy. The greater the number, the higher the tolerance limit. The choice of tolerance is guided by the functional requirements of the product and economy of manufacture. The degree of accuracy attained depends on the type and condition of the machine tool used.

Tolerances grades for different applications

Fundamental tolerance	Applications
IT01–IT4	For production of gauges, plug gauges, and measuring instruments
IT5–IT7	For fits in precision engineering applications such as ball bearings, grinding, fine boring, high-quality turning, and broaching
IT8–IT11	For general engineering, namely turning, boring, milling, planning, rolling, extrusion, drilling, and precision tube drawing
IT12–IT14	For sheet metal working or press working
IT15–IT16	For processes such as casting, stamping, rubber moulding, general cutting work, and flame cutting

Tolerance values corresponding to grades IT5–IT16 are determined using the standard tolerance unit (i, in μm), which is a function of basic size.

$$i = 0.453 \sqrt[3]{D} + 0.001D \text{ microns}$$

where D is the diameter of the part in mm. The linear factor 0.001D counteracts the effect of measuring inaccuracies that increase by increasing the measuring diameter. By using this formula, the value of tolerance unit ‘i’ is obtained for sizes up to 500 mm.

D is the geometric mean of the lower and upper diameters of a particular diameter step within which the given or chosen diameter D lies and is calculated by using the following equation:

$$\sqrt{D_{\text{max}} \times D_{\text{min}}}$$

The various steps specified for the diameter steps are as follows:

1–3, 3–6, 6–10, 10–18, 18–30, 30–50, 50–80, 80–120, 120–180, 180–250, 250–315, 315– 400, 400–500 500–630, 630–800, and 800–1000 mm.

Tolerances have a parabolic relationship with the size of the products. The tolerance within which a part can be manufactured also increases as the size increases. The standard tolerances corresponding to IT01, IT0, and IT1 are calculated using the following formulae:

$$\mathrm{IT01}: 0.3 + 0.008D$$

$$\mathrm{IT0}: 0.5 + 0.012D$$

$$\mathrm{IT1}: 0.8 + 0.020D$$

The values of tolerance grades IT2–IT4, which are placed between the tolerance grades of IT1 and IT5, follow a geometric progression to allow for the expansion and deformation affecting both the gauges and the workpieces as dimensions increase. For the tolerance grades IT6–IT16, each grade increases by about 60% from the previous one, as in table below .

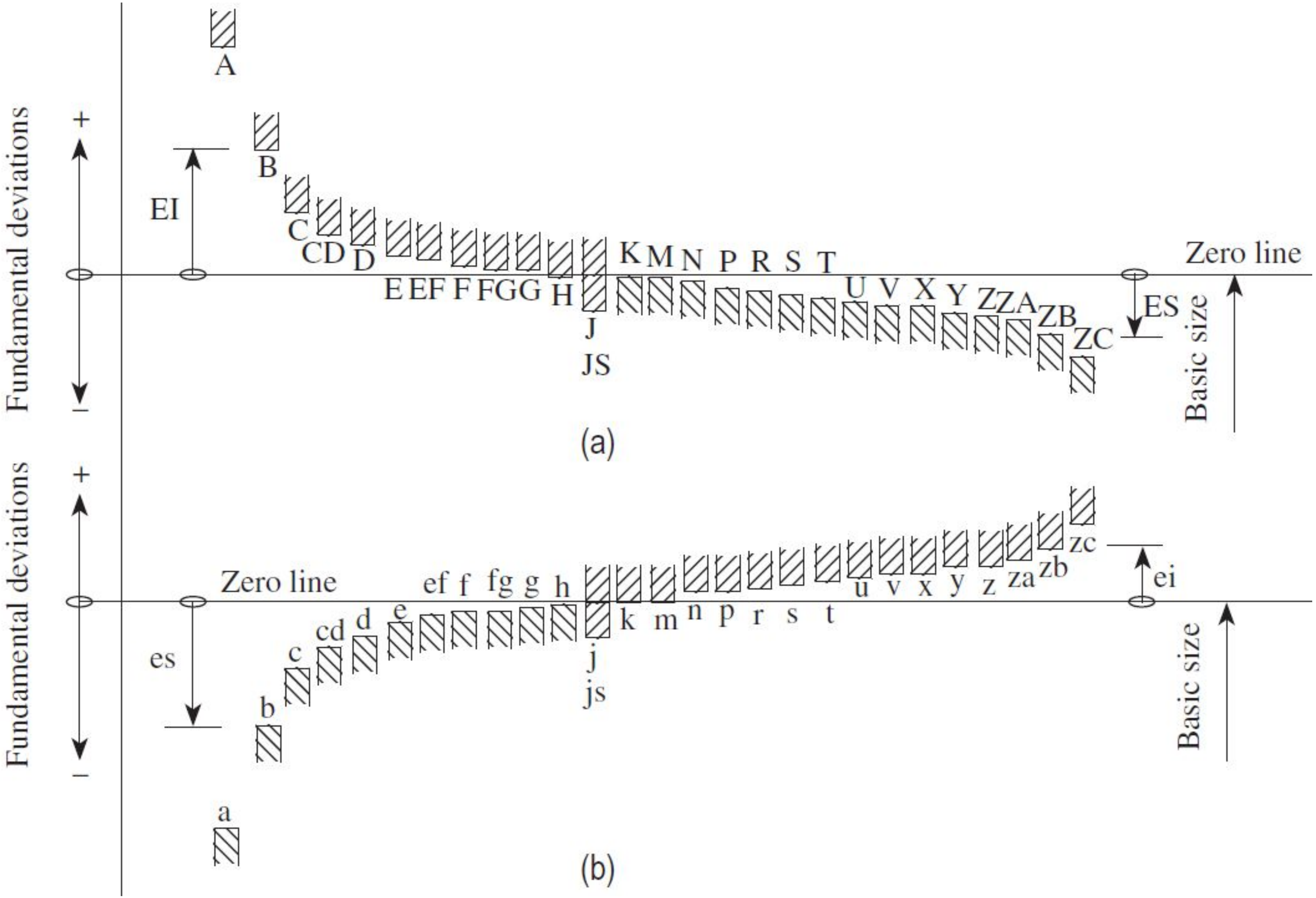
Tolerance grade	IT6	IT7	IT8	IT9	IT10	IT11	IT12	IT13	IT14	IT15	IT16
Standard tolerance unit (*i*)	10	16	25	40	64	100	160	250	400	640	1000

The tolerance zone is governed by two limits: the size of the component and its position related to the basic size. The position of the tolerance zone, from the zero line (basic size), is determined by fundamental deviation.

The values of these tolerance grades or fundamental deviations depend on the basic size of the assembly. The different values of standard tolerances and fundamental deviations can be obtained by referring to the design handbook. The choice of the tolerance grade is governed by the type of manufacturing process and the cost associated with it.

From Fig 1 ., a typical case can be observed in which the fundamental deviation for both hole H and shaft h having a unilateral tolerance of a specified IT grade is zero.

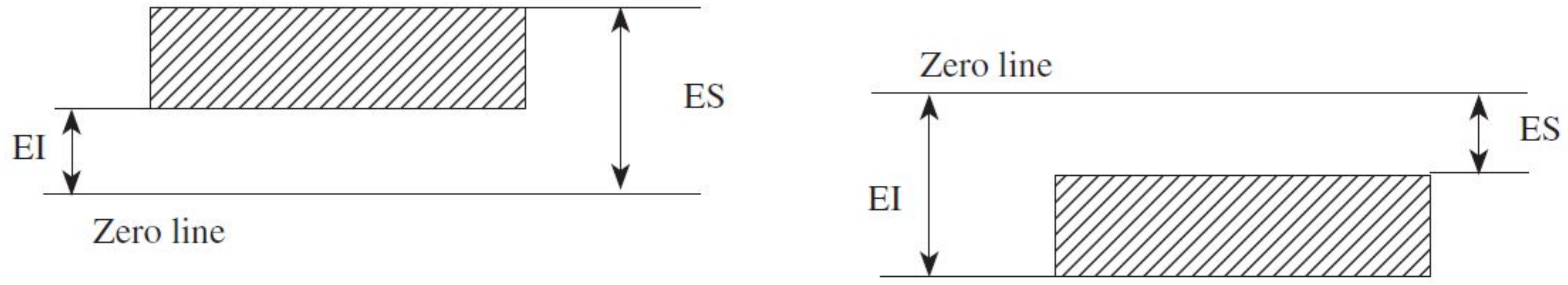
The first eight designations from A (a) to H (h) for holes (shafts) are intended to be used in clearance fit, whereas the remaining designations, JS (js) to ZC (zc) for holes (shafts), are used in interference or transition fits. For JS, the two deviations are equal and given by ±IT/2.


Fundamental deviations
+
-
EI
A
B
C
CD
D
E EF F FG G H
J
JS
K M N P R S T
U V X Y Z ZA ZB ZC
Zero line
ES
Basic size
(a)
Fundamental deviations
+
-
es
Zero line
a
b
c
cd
d
e
ef f fg g h
j
js
k m n p r s t
u v x y z za zb zc
ei
Basic size
(b)

Consider the designation 40 H7/d9. In this example, the basic size of the hole and shaft is 40 mm. The nature of fit for the hole basis system is designated by H and the fundamental deviation of the hole is zero. The tolerance grade is indicated by IT7. The shaft has a d-type fit for which the fundamental deviation (upper deviation) has a negative value, that is, its dimension falls below the basic size having IT9 tolerance.

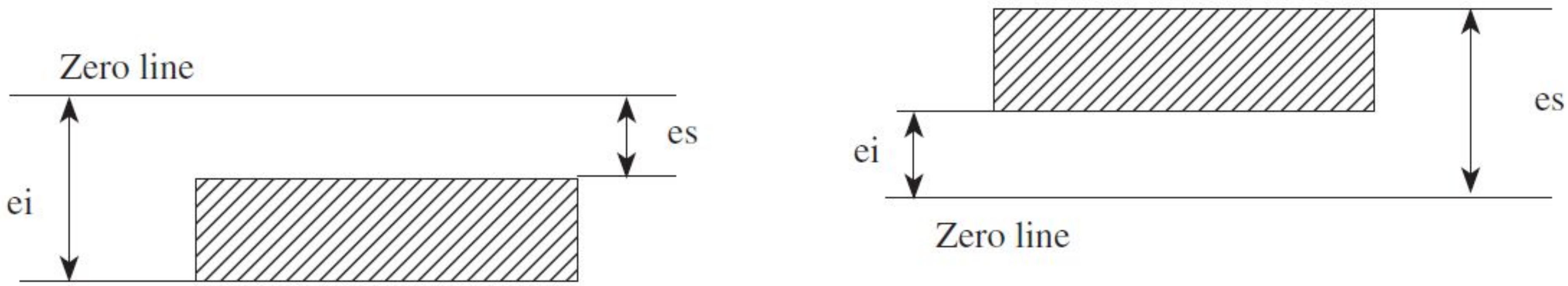
Depending on the application, numerous fits ranging from extreme clearance to extreme interference can be selected using a suitable combination of fundamental deviations and fundamental tolerances. From Fig, it can be seen that the lower deviation for the holes ‘A’ to ‘G’ is above the zero line and that for ‘K’ to ‘ZC’ is below the zero line. In addition, it can be observed that for shafts ‘a’ to ‘g’, the upper deviation falls below the zero line, and for ‘k’ to ‘zc’ it is above the zero line.

It can be seen from Fig. 3.17 that ‘EI’ is above the zero line for holes ‘A’ to ‘G’, indicating positive fundamental deviation. In contrast, Fig. 3.18 shows that ‘ei’ is below the zero line for the shafts ‘a’ to ‘g’ and therefore the fundamental deviation is negative. In addition, from Figs 3.17 and 3.18, it can be observed that for holes ‘K’ to ‘ZC’, the fundamental deviation is negative (‘EI’ below the zero line), whereas for shafts ‘k’ to ‘zc’, it is positive (‘ei’ above the zero line).



(a) (b)

Fig. 3.17 Deviations for holes (a) Deviations for A to G (b) Deviations for K to Z_i



(a) (b)

Fig. 3.18 Deviations for shafts (a) Deviations for a to g (b) Deviations for k to Z_i

It follows from Figs 3 .17 and 3 .18 that the values of 'ES' and 'EI' for the holes and 'es' and 'ei' for the shafts can be determined by adding and subtracting the fundamental tolerances, respectively. Magnitude and sign of fundamental deviations for the shafts, either upper deviation 'es' or lower deviation 'ei' for each symbol, can be determined from the empirical relationships listed in Tables 3.5 and 3.6, except for shafts j and js for which there is no fundamental deviation. The values given in Tables 3.5 and 3.6 are as per IS: 919. The other deviations of the holes and shafts can be determined using the following relationships that can be derived from Figs 3.17 and 3.18

1. For holes A to G, EI is a positive fundamental deviation and EI = ES − IT.
2. For holes K to Z*i*, the fundamental deviation ES is negative and ES = EI − IT.
3. For shafts ‘a’ to ‘g’, the fundamental deviation es is negative and es = ei − IT.
4. For shafts ‘k’ to ‘Z*i*’, the fundamental deviation ei is positive, and ei = es − IT.

Table 3.5 Fundamental deviation formulae for shafts of sizes up to 500 mm

Upper deviation (es)		**Lower deviation (ei)**	
Shaft designation	In μm (for D in mm)	Shaft designation	In μm (for D in mm)
a	$= -(265 + 1.3D)$ for $D \le 120$	j5–j8	No formula
		(For js two deviations are equal to IT ± 2)	
	$= -3.5D$ for $D > 120$	k4–k7	$= +0.6\sqrt[3]{D}$
		For grades ≤k3 to ≥k8	$= 0$
b	$= -(140 + 0.85D)$ for $D \le 160$	m	$= +(IT7 - IT6)$
	$= -1.8D$ for $D > 160$	n	$= +5D^{0.34}$
		p	$= +$ IT7 + 0 to 5
c	$= -52D^{0.2}$ for D + IT7 + 0 to 40	r	= Geometric mean of ei values for p and s
	$= -(95 + 0.8D)$ for $D > 40$		$= +$IT8 + 1 to 4 for $D \le 50$
d	$= -16D^{0.44}$	s	$= +IT7 + 0.4D$ for $D > 50$
e	$= -11D^{0.41}$	t	$= +IT7 + 0.63D$
f	$= -5.5D^{0.41}$	u	$= +IT7 + D$
g	$= -2.5D^{0.34}$	v	$= +IT7 + 1.25D$

Production of large-sized components is associated with problems pertaining to manufacture and measurement, which do not exist in the case of smaller-sized components. As the size of the components to be manufactured increases, the difficulty in making accurate measurements also increases. Variations in temperature also affect the quality of measurement.

When the size of the components to be manufactured exceeds 500 mm, the tolerance grades IT01–IT5 are not provided, as they are considered to be too small. The fundamental tolerance unit in case of sizes exceeding 500 and up to 3150 mm is determined as follows:

$i = 0.004D + 2.1D$, where D is 0.001 mm.

The fundamental deviations for holes and shafts are given in Table 3.7, which are as per IS: 2101.(Fundamental deviation for shafts and holes of sizes from above 500 to 3150 mm)

Shafts			Holes			Formula for deviations in μm
Type	Fundamental deviation	Sign	Type	Fundamental deviation	Sign	(for D in mm)
d	es	–	D	EI	+	$16D^{0.44}$
e	es	–	E	EI	+	$11D^{0.41}$
f	es	–	F	EI	+	$5.5D^{0.41}$
g	es	–	G	EI	+	$2.5D^{0.34}$
h	es	No sign	H	EI	No sign	0
js	ei	–	JS	ES	+	$0.5\mathrm{IT}\pi$
k	ei	–	K	ES	–	0
m	ei	+	M	ES	–	$0.024D + 12.6$
n	ei	+	N	ES	–	$0.04D + 21$
P	ei	+	P	ES	–	$0.072D + 37.8$
r	ei	+	R	ES	–	Geometric mean of the values for p and s or P and S
s	ei	+	S	ES	–	$\mathrm{IT}7 + 0.4D$
t	ei	+	T	ES	–	$\mathrm{IT}7 + 0.63D$
u	ei	+	U	ES	–	$\mathrm{IT}7 + D$

For the following hole and shaft assembly, determine (a) hole and shaft tolerance and (b) type of fit

$$\text{Hole} = 20^{+0.05}_{+0.00}\ \text{mm and shaft} = 20^{+0.08}_{+0.06}\ \text{mm}$$

Illustration for Determining Type of Fit

Determine type of Fit of 55 H7 f8

Determine the dimensions and tolerances of shaft and hole having size of 30 H7 / h8 fit. Also determine the allowance (i.e. minimum clearance) and maximum clearance.

- For hole H 7 and shaft h 8, the fundamental deviation is zero.

As 30 mm lies in the range of 18 and 30 mm , D = $\sqrt{18 \, X \, 30}$ = 23.2 mm

i = $0.45 \, (\sqrt[3]{D})$ + 0.001 D

= 0.45 $\sqrt[3]{23.2}$ +0.001 x 23.2 microns

= 0.45 X 2.855 + 0.023 = 1.308 microns

IT 7 = 16 i = 16 X 1.308 microns = $\frac{16 \, X \, 1.308}{1000}$ mm ≈ 0.0020928 ≈ 0.021 mm

Min. size of hole = 30.0000 mm   Max size of hole = 30.021 mm

Max. size of shaft = 30.000 mm

value of IT8 for shaft = 25 i = 25 x 1.308 = 32.7 microns= 0.0033 mm

min. size of shaft = 30.000 - 0.0033 = 29.9967 mm

Allowance or minimum clearance = =Min.size of hole -Max. size of shaft= 30.0000- 30.000 = 0

Max. clearance = Max. size of hole - Min. size of shaft = = 30.0021 - 29.2967 = 0.0054 mm